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Publications and Preprints

[1] **O. Matsyshyn**, L.-k. Shi, I. Sodemann. **High-Temperature Quantum Oscillations of a Non-equilibrium Non-Fermi Liquid**. *arXiv:2512.21763*, PRB accepted.

[2] J. Tan†, **O. Matsyshyn**†, G. Vignale, J. C. W. Song. **Nonequilibrium Exchange Nonlinear Hall Effect**. *arXiv:2512.06074*. †Equal contribution.

[3] **O. Matsyshyn**, J. C. W. Song. **Superconducting Photocurrent and Light-Enriched Supercurrent Phase Relation**. *arXiv:2507.06290*.

[4] F. J. Garc'ia de Abajo, *et al.*, **O. Matsyshyn**, *et al.* **Roadmap for Photonics with 2D Materials**. *ACS Photonics* 2025, 5c00353.

[5] **O. Matsyshyn**, G. Vignale, J. C. W. Song. **Superconducting Berry Curvature Dipole**. *Physical Review Letters* 136, 246902.

[6] L.-k. Shi†, **O. Matsyshyn**†, J. C. W. Song, I. Sodemann. **The Ultra-critical Floquet Non-Fermi Liquid**. *Physical Review Letters* 134, 196401 (2025). †Equal contribution.

[7] **O. Matsyshyn**, Y. Xiong, J. C. W. Song. **Kramers dichroism in PT-symmetric magnets**. *Phys. Rev. Research 8, 033046*.

[8] L.-k. Shi†, **O. Matsyshyn**†, J. C. W. Song, I. Sodemann Villadiego. **Floquet Fermi Liquid**. *Physical Review Letters* 132, 146402 (2024). Selected for the cover. †Equal contribution.

[9] J.-X. Hu, **O. Matsyshyn**, J. C. W. Song. **Nonlinear Superconducting Magnetoelectric Effect**. *Physical Review Letters* 134, 026001 (2025).

[10] **O. Matsyshyn**, J. C. W. Song, I. Sodemann Villadiego, L.-k. Shi. **The Fermi-Dirac Staircase Occupation of Floquet Bands and Current Rectification Inside the Optical Gap of Metals**. *Physical Review B* 107, 195135 (2023). Editorial Suggestion.

[11] **O. Matsyshyn**, Y. Xiong, A. Arora, J. C. W. Song. **Layer Photovoltaic Effect in van der Waals Heterostructures**. *Physical Review B* 107, 205306 (2023).

[12] L.-k. Shi, **O. Matsyshyn**, J. C. W. Song, I. Sodemann Villadiego. **The Berry Dipole Photovoltaic Demon and the Thermodynamics of Photo-current Generation Within the Optical Gap of Metals**. *Physical Review B* 107, 125151 (2022). Editorial Suggestion.

[13] **O. Matsyshyn**, F. Piazza, R. Moessner, I. Sodemann. **The Rabi Regime of Current Rectification in Solids**. *Physical Review Letters* 127, 126604 (2021).

[14] **O. Matsyshyn**, U. Dey, I. Sodemann, Y. Sun. **The Berry Phase Rectification Tensor and the Solar Rectification Vector**. *Journal of Physics D: Applied Physics* 54, 404001 (2021).

[15] **O. Matsyshyn**, I. Sodemann. **The Non-linear Hall Acceleration and the Quantum Rectification Sum Rule**. *Physical Review Letters* 123, 246602 (2019). Cover article.

[16] **O. Matsyshyn**, A. I. Yakimenko, E. V. Gorbar, S. I. Vilchinskii, V. V. Cheianov. **p-wave Superfluidity in Mixtures of Ultracold Fermi and Spinor Bose Gases**. *Physical Review A* 98, 043620 (2018).

[17] A. I. Yakimenko, **O. Matsyshyn**, A. O. Oliinyk, V. M. Biloshytskyi, O. G. Chelpanova, S. I. Vilchynskii. **Analogues of Josephson Junctions and Black Hole Event Horizons in Atomic Bose–Einstein Condensates**. *Low Temperature Physics* 44, 1032 (2018). Cover article.

Papers in preparation

[1] **O. Matsyshyn**, Valla Fatemi, Justin Song. **Superconducting Berry Curvature Dipole Qubit**.

[2] Li-kun Shi, **O. Matsyshyn**, Justin C. W. Song, and Inti Sodemann Villadiego. **Bosonic Floquet Fermi Surfaces**.

[3] Zumeng Huang, **O. Matsyshyn**, et al. **Néel-Controlled Kinetic Scar Photocurrents from an Incoherent Cascade**.

Book and Public Outreach

- **Toys or Physics? Explaining Physics Through Toys** *2023*
  World Scientific; DOI: 10.1142/13547
  - Educational book introducing core physics concepts through everyday toys and hands-on activities.
  - Awarded the 2024 Nautilus Silver Medal in the Middle Grade Non-Fiction category.